Base paper. F. N. Nagoor Pitchai, E. J. Tanner, N. Khetan *et al.*, *Engineered deletions of HIV replicate conditionally to reduce disease in nonhuman primates*, Science **385**, eadn5866 (2024). Model inherited from L. S. Weinberger, D. V. Schaffer, A. P. Arkin, *J. Virol.* **77**, 10028 (2003).

The problem in one line. The TIP mobilises only from dually infected cells, so its reproduction scales with the wild-type load it is suppressing. It starves its own source, and the system runs to coexistence with a floor viral load below which the TIP goes extinct. In the published model that floor is $V_H^{\text{floor}} = d/[\beta_H(A-1)]$, fixed by d, the loss rate of a TIP-only cell. The design changes that one rate: transduced cells are partitioned into the central-memory pool, where an integrated provirus is copied with the genome and no Gag is required. The numerator becomes the *net* loss rate of that compartment, $\mu = d_M - \rho(1-\varepsilon)$, which can be driven small without touching conditional replication, the packaging advantage, or the safety argument.

1. Specific Aims

Aim 1. A value for p, the probability that a TIP-only cell is superinfected before it is lost, at human chronic set point, and the implied R_0^{TIP} for the published TIP. Deliverable: one number with an interval, and a verdict on whether the published construct sits above or at $R_0^{\text{TIP}} = 1$ in humans.

Approach. The Discussion of the base paper cites 5 to 10% of infected cells carrying multiple proviruses at chronic set point. That is a prevalence conditioned on infection at one instant, not the cumulative probability that a single TIP-only cell is superinfected over its life; the two differ by the ratio of compartment lifetimes and by how the infected pool is sampled. I convert one into the other using the TIP-only cell lifetime $1/(d+\beta_H V_H)$ and $\beta_H V_H$ at the model's own HIV equilibrium. The threshold is $p^* = 1/A$, independent of d, and equals 8.3% for $A = 12$. Both numbers are already in the paper, so this check needs no simulation. $A = 12$ is itself a ceiling: the Fig. 2F three-colour coculture starts from producer cells already transduced and then infected, so superinfection is arranged rather than sampled.

Aim 2. The seven-state model with the maintenance term, implemented and verified, and the persistence boundary in the (μ, A) plane. Deliverable: a runnable solver, the threshold condition derived in §3, and the region of (μ, A) that places the floor at or below 10^3 copies mL^{-1}.

Approach. Add $\rho(1-\varepsilon)I_T$ to the TIP-only equation and replace the TIP-only loss rate d by the central-memory loss rate d_M; everything else is left as published. Linearising about the HIV-only equilibrium gives the boundary in closed form, so the numerics are a check rather than the argument. Verification first: with $\rho = 0$ and $d_M = d$ the code must reproduce the separation the base paper's Fig. 4D rests on, the basic model failing the treated animals on the peak-to-set-point ratio in $\log_{10}$ while the TIP extension fits both.

Aim 3. A bound on ε, the fitness cost of carrying the TIP in a central-memory cell, and the measurement that would produce it. Deliverable: the admissible interval for ε given published memory CD4^+ kinetics, and a protocol that separates the intrinsic cost from the protection the TIP confers against HIV killing.

Approach. ε is the one parameter in the extended model that no published experiment bounds. Every comparison of TIP-transduced against control cells in the base paper (fig. S16E, fig. S6BC) is run in the presence of HIV, so the intrinsic cost and the protection from HIV killing enter the same number with opposite signs and neither is recoverable from it. The measurement has to be run without HIV: transduced and untransduced central-memory cells from one donor, same well, barcode ratio followed across divisions. The anchor on what is achievable is the latent reservoir, the same compartment carrying an integrated provirus, which decays with $t_{1/2} = 44.2$ months.

2. The model

Six states, per mL of blood: targets T, HIV-infected I_H, TIP-only I_T, dually infected I_D, free HIV V_H, free TIP V_T. The equations are the TIP extension of Weinberger *et al.* (2003) as used by the base paper, with two changes, both in the I_T equation: the added self-renewal term, and the loss rate d_M of the central-memory pool in place of the loss rate d of a short-lived target cell.

$$
\begin{aligned}
\dot{T} &= \lambda - dT - \beta_H V_H T - \beta_T V_T T, & \dot{I}_D &= \beta_T V_T I_H + \beta_H V_H I_T - \delta_D I_D,\\
\dot{I}_H &= \beta_H V_H T - \delta I_H - \beta_T V_T I_H, & \dot{V}_H &= N\delta I_H + (1-D)\,N\delta_D I_D - cV_H,\\
\dot{I}_T &= \beta_T V_T T + \underbrace{\rho(1-\varepsilon)I_T}_{\text{added}} - d_M I_T - \beta_H V_H I_T, & \dot{V}_T &= \Pi N \delta_D I_D - cV_T.
\end{aligned}
$$

D is the inhibition of wild-type output from a dually infected cell, so that a fraction $1-D$ survives; this is $1-D_{\mathrm{W}}$ in the notation of Weinberger *et al.*, where their optimum $D_{\mathrm{W}} \approx 0.2$ is 80% inhibition. Π is the packaging advantage, equal to P^2 in their notation since two genomes are packaged per virion. Conditional replication is the absent production term in I_T: a TIP-only cell makes nothing.

	Meaning	Value	Units	Source
λ	target cell production	3.1×10^4	cells $\mathrm{mL}^{-1}\mathrm{d}^{-1}$	Weinberger 2003, Table 1 (31 cells $\mu\mathrm{L}^{-1}\mathrm{d}^{-1}$)
d	loss rate, uninfected target	0.02	d^{-1}	Weinberger 2003, Table 1
β_H	infection rate per virion	1.875×10^{-7}	mL $\mathrm{virion}^{-1}\mathrm{d}^{-1}$	Weinberger 2003, Table 1 (k, volume units reconciled so $T^* = c/N\beta_H$ returns the tabulated 800 cells $\mu\mathrm{L}^{-1}$)
β_T	TIP infection rate	$= \beta_H$	mL $\mathrm{virion}^{-1}\mathrm{d}^{-1}$	Weinberger 2003, Theory: the TIP is phenotypically identical and infects at the same per-contact rate
δ	loss rate, I_H	0.7	d^{-1}	Weinberger 2003, Table 1
δ_D	loss rate, I_D	$(1-D)\delta = 0.14$	d^{-1}	Weinberger 2003, Table 1 ($\delta' = D_{\mathrm{W}}\delta$); cancels from the threshold
N	burst size	200	virions cell^{-1}	Weinberger 2003, Table 1
c	virion clearance	30	d^{-1}	Weinberger 2003, Table 1 (3 d^{-1} in their no-eclipse variant, footnote a)
D	inhibition of HIV output from I_D	0.8	–	Weinberger 2003, Fig. 3b, optimum $D_{\mathrm{W}} \approx 0.2$; to be replaced by the measured interference of Pitchai 2024, Fig. 2H
Π	packaging advantage	12	–	set equal to A below; $\Pi = P^2$ gives $P \approx 3.5$, inside the 1–100 range of Weinberger 2003, Table 1
A	TIP output per superinfection	12 ± 0.5	–	Pitchai 2024, Fig. 2F (R_0^{TIP} by three-colour coculture), read as A because that assay arranges superinfection
ρ	self-renewal, central memory	0.015	d^{-1}	Macallan *et al.* 2004, *J. Exp. Med.* **200**:255, Table II: 1.5%/day for $\mathrm{CD45R0}^+\mathrm{CCR7}^+$ $\mathrm{CD4}^+$
d_M	loss rate, central memory	$= \rho$	d^{-1}	**assumption**: the untransduced compartment is at steady state and self-renewing, so division balances loss
ε	fitness cost of carrying the TIP	unknown	–	**unmeasured**; the object of Aim 3
T^*	$\mathrm{CD4}^+$ set point, pre-therapy	8×10^5	cells mL^{-1}	Weinberger 2003, Table 1 (800 cells $\mu\mathrm{L}^{-1}$)
V_0	HIV set point, pre-therapy	10^5	copies mL^{-1}	Weinberger 2003, Table 1
V_{target}	set point the design must hold	10^3	copies mL^{-1}	**assumption**: one $\log_{10}$ below the SHIV set point reached under TIP, above the 400 copies mL^{-1} failure threshold cited in Weinberger 2003
μ_{ref}	net decay, latent reservoir	5.2×10^{-4}	d^{-1}	Siliciano *et al.* 2003, *Nat. Med.* **9**:727, $t_{1/2} = 44.2$ months

One unit convention, labelled as an assumption: V is carried in virions mL^{-1} and viral loads reported as RNA copies mL^{-1} are read as 1 copy = 1 virion, as in Weinberger 2003, Table 1. Two derived quantities are used below. At the HIV-only equilibrium $T^* = c/(N\beta_H)$, so the TIP's output per superinfection, $A = \Pi N \beta_T T^*/c$, reduces to $A = \Pi\,\beta_T/\beta_H = \Pi$. And $R_0^{\mathrm{TIP}} = pA$ with $p = \beta_H V_H/(d_M + \beta_H V_H - \rho(1-\varepsilon))$.

3. The one quantity that has to come out right

$\mu = d_M - \rho(1-\varepsilon)$, the net loss rate of the TIP-carrying central-memory compartment, in d^{-1}.

Linearising the three TIP equations about the HIV-only equilibrium, and keeping only the route $I_T \to I_D \to V_T \to I_T$, the TIP invades and persists if and only if

$$(A-1)\,\beta_H V_H \;>\; \mu, \qquad \text{equivalently} \qquad V_H^{\mathrm{floor}} \;=\; \frac{\mu}{\beta_H (A-1)}.$$

With $\rho = 0$ and $d_M = d$ this returns the published floor, so the design enters through μ alone. Dropping the second route, $V_T \to I_H \to I_D$, only raises R_0^{TIP}, so the bound below is conservative.

The value it has to reach. For the TIP to persist while holding the load at $V_{\mathrm{target}} = 10^3$ copies mL^{-1} with $A = 12$,

$$\mu \;\le\; (A-1)\,\beta_H V_{\mathrm{target}} \;=\; 11 \times 1.875 \times 10^{-7} \times 10^3 \;=\; 2.1 \times 10^{-3}\ \mathrm{d}^{-1},$$

a net half-life of the compartment of at least 336 days, about 11 months. Under the labelled assumption $d_M = \rho$ this is $\mu = \rho\varepsilon$, so the bar on the unmeasured parameter is $\varepsilon \le 13.7\%$. Two reference points fix the scale: the published model's TIP-only loss rate is $d = 0.02\ \mathrm{d}^{-1}$, an order of magnitude outside the bar, and the latent reservoir in the same compartment sits at $\mu_{\mathrm{ref}} = 5.2 \times 10^{-4}\ \mathrm{d}^{-1}$, a factor of four inside it.

M3 reports μ, and the implied $\varepsilon_{\mathrm{crit}}$, against the value above.